

Japanese and Chinese Language Learning in Alternate Realities: How Effective are AR and VR Technologies?

7-papers assignment

Japanese and Chinese are among some of the hardest languages for native English speakers to learn. Not only are the sounds that you have to make quite foreign from those found in English, but the writing systems are also completely different and require long, intensive study (Kanev et al., 2015; Wang and Wang, 2024). While traditional study methods such as using a textbook or taking in-person language classes have been proven to work, current developments in technology allow for alternate approaches to learning these languages. Namely, augmented reality (AR) and virtual reality (VR) study tools are seeing more utilization in language learning environments. The main topic of this overview, therefore, is how effective these tools are at supporting Japanese and Chinese language learning compared with traditional study methods.

Increased learning performance

In research focusing on how effective AR technology is in teaching Japanese compound verbs to non-Japanese students, Geng and Yamada (2020) utilized an AR application that could scan verbs on physical cards and play an animation acting out that verb. One group of students would then be instructed to use the application and combine two cards to find the correct compound verbs. They also received a paper handout explaining the meaning of these verbs. At the same time, another group of students studied the same verbs using a paper-based image schema (an abstract representation of the meaning of the verb) in addition to the same handout as the first group. After a post-test, results showed that the students who used the AR application showed better retention of the material studied. They were particularly more adept at recognizing correct compound verb combinations, due to the fact that the AR application allowed them to physically interact with the study material. In another research, Geng and Yamada (2023) improved on their application by including written explanations of the verbs on the screen, which further increased learning performance as the users of the application no longer had to switch their focus to the paper handout.

In a similar study utilizing scannable AR cards but focused on Chinese language learning, Chen (2024) also found benefits to using the technology. After scanning a card, a video would pop up and explain the character alongside an animation of the character's evolution throughout history. This multimedia aspect was shown to improve memorization. Yet another research paper on AR technology utilization for Chinese character acquisition by Wang and Wang (2024) shows that memory retention of the meanings and stroke orders of these characters in particular was significantly higher by users of AR than those who used traditional study methods. This is in part explained by the animated and interactive features provided by the AR technology, which make it more engaging to learners. However, Wang and Wang (2024) also

admit that the assistance by AR technology alone does not guarantee actual acquisition of Chinese characters and that students will still have to embrace traditional study methods such as repetition and imitation in order to really become familiar with the characters.

Finally, Chen and Yuan (2023) compared the effectiveness of VR study tools with traditional study methods regarding Chinese vocabulary learning. With the same end goal of learning 30 words, one group of students had two 30-minute study sessions using an Oculus Quest 2, while the other group had similar-length study sessions in a traditional classroom environment. Results showed that, while pre-test scores were nearly identical between the groups, the VR group had substantially higher scores than the classroom group in both a post-test directly after the study sessions and a delayed test 14 days after the experiment. This was accredited to the immersive contexts made available by the VR tool and the fact that students received instant feedback in response to wrong answers, allowing them to quickly learn from their mistakes.

Motivated and interested students

Another recurring theme in research on AR and VR tools in language learning environments is the increased motivation for studying and high interest in using these tools among students. Kanev et al. (2015), who frame their AR tools as games and augmented toys, noted high levels of interest in their alternate ways of studying Japanese kanji characters (similar logographic characters as Chinese) utilizing AR over traditional methods. Furthermore, respondents answered that they thought they would learn better through these methods.

Xie et al. (2019) had students of the Chinese language use Google Expedition, a VR teaching tool, to do presentations in Chinese, virtually guiding their fellow students through various famous locations in China. The usage of this immersive tool over more traditional presentation tools such as Microsoft PowerPoint increased motivation and engagement among the students, as it felt as if they were really at the places being talked about instead of having to imagine it. Additionally, some of the students reported feeling less anxiety speaking in Chinese while using Google Expeditions, stating that the VR tool could distract the listeners, giving the speaker some more time to think about what words they wanted to say.

Remaining issues

While there seem to be many advantages to using AR and VR technology in language learning environments, some potential obstacles have also been acknowledged. For one, while some students are enthusiastic about using these tools more in the future (Wang and Wang, 2024), others cited technical issues hindering their studying and even reported dizziness using VR devices in class, limiting their usage of the tools (Xie et al. 2019). Finally, the cost of VR tools in particular appears to be a deterrent as well. Some students who had a positive experience using VR tools for studying reported that they felt the required devices were unaffordable for them.

References:

Chen, C., & Yuan, Y. (2023). Effectiveness of Virtual Reality on Chinese as a second language vocabulary learning: Perceptions from international students. *Computer Assisted Language Learning*.

Chen, Y. (2024). The effect of applying augmented reality to Chinese character learning on novice CFL learners. *Education and Information Technologies*.

Geng, X., & Yamada, M. (2020). An augmented reality learning system for Japanese compound verbs: Study of learning performance and cognitive load. *Smart Learning Environments*, 7(27).

Geng, X., & Yamada, M. (2023). Using learning analytics to investigate learning processes and behavioural patterns in an augmented reality language learning environment. *Journal of Computer Assisted Learning*, 39(2), 532–546.

Kanev, K., Oido, I., Hung, P. C. K., Kapralos, B., & Jenkin, M. (2015). Case Study: Approaching the Learning of Kanji Through Augmented Toys in Japan. In P. C. K. Hung (Ed.), *Mobile Services for Toy Computing* (pp. 175–192). Springer International Publishing.

Wang, Y., & Wang, X. (2024). Using AR-assisted approach in Chinese character acquisition for adult CFL learners. *Education and Information Technologies*.

Xie, Y., Ryder, L., & Chen, Y. (2019). Using Interactive Virtual Reality Tools in an Advanced Chinese Language Class: A Case Study. *TechTrends*, 63, 251–259.